THE INCLUSION OF ADEQUATE MODULES HAS A RIGHT ADJOINT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's footnote naming the right adjoint 'the adequator' is part of the delivered statement and is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Modules on the big site may be studied on the category of affines. Topologies, Lemma topologies-lemma-affine-big-site-fppf, invoked at proof lines 2488-2491. Lets the argument be carried out in terms of module-valued functors on Alg_A.
- The construction Q is a right adjoint to the inclusion of adequate functors. Same chapter, Lemma lemma-adjoint, invoked at proof lines 2495-2496. Provides the functor appearing in the definition of the adequator.
- The identifications of the two vertical arrows of the displayed square. Same chapter, Lemma lemma-adequate-affine and Section section-quasi-coherent, invoked at proof lines 2509-2512. Justify the square in which the argument takes place.

Source reference. Original lemma 06ZJ: sources/stacks-adequate/pinned-original-source.tex, source lines 1738–1738. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06Z6: sources/stacks-adequate/pinned-original-source.tex, source lines 1043–1043. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 06US: sources/stacks-adequate/pinned-original-source.tex, source lines 321–321. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 021V: sources/stacks-chow/tags.txt, source lines 2684–2684. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $U = \mathrm{Spec}(A)$ be an affine scheme. The inclusion functor*

$$Adeq(\mathcal{O}) \to Mod((Sch/U)_\tau, \mathcal{O})$$

has a right adjoint A[1]. Moreover, the adjunction mapping $A(\mathcal{F}) \to \mathcal{F}$ is an isomorphism for every adequate module $\mathcal{F}$.

Proof. By Topologies, Lemma 021V (and similarly for the other topologies) we may work with $\mathcal{O}$-modules on $(Aff/U)_\tau$. Denote $\mathcal{P}$ the category of module-valued functors on Alg_A and $\mathcal{A}$ the category of adequate functors on Alg_A. Denote $i : \mathcal{A} \to \mathcal{P}$ the inclusion functor. Denote $Q : \mathcal{P} \to \mathcal{A}$ the construction of Lemma 06Z6. We have the commutative diagram

$$\begin{array}{ccccc} Adeq(\mathcal{O}) & \xrightarrow[k]{} & Mod((Aff/U)_\tau, \mathcal{O}) & \xrightarrow[j]{} & PMod((Aff/U)_\tau, \mathcal{O}) \\ \| & & & & \| \\ \mathcal{A} & & \xrightarrow{\qquad\qquad i \qquad\qquad} & & \mathcal{P} \end{array} \tag{0.1.1}$$

The left vertical equality is Lemma 06ZJ and the right vertical equality was explained in Section 06US. Define $A(\mathcal{F}) = Q(j(\mathcal{F}))$. Since j is fully faithful it follows immediately that A is a right adjoint of the inclusion functor k. Also, since k is fully faithful too, the final assertion follows formally. □

[1]This is the "adequator".